Fall 2026 ECE 241: Digital Design (Units: 1.5)

Territory Acknowledgement

We acknowledge and respect the Lək̓ʷəŋən (Songhees and Xʷsepsəm/Esquimalt) Peoples on whose territory the university stands, and the Lək̓ʷəŋən and W̱SÁNEĆ Peoples whose historical relationships with the land continue to this day.

Withdrawal without reduction of tuition fees: 2026-09-22
Withdrawal with 50% reduction of tuition fees: 2026-10-13
Last day for withdrawal (no fees returned): 2026-10-31

TA Information

TA Name	E-mail	Section
TBD	TBD	TBD

Course Pre- & Co-requisites

Minimum second-year standing.

Course Objectives

- To understand fundamental concepts in analyzing and designing combinational and sequential logic circuits.
- To gain experience with implementation and testing of digital systems.

Learning Outcomes

- Understanding Boolean algebra and its applications.
- Understanding binary and hexadecimal number systems.
- Understanding the characteristics and behavior of logic gates.
- Acquiring the capability to analyze and design combinational and sequential logic circuits.
- Gaining an introductory experience with implementation and testing of digital logic systems with an emphasis on synthesis and analysis of HDL designs using modern EDA tools.

Syllabus

Binary numbers. Boolean algebra and logic gates. Combinational circuits and gate-level minimization. Synchronous sequential logic circuits. Finite state machines. Registers and counters. Implementation technologies.

Textbook & Reference Materials

Required Text	
Title:	Digital Design With An Introduction to the Verilog HDL, VHDL and SystemVerilog (6th Edition)
Author:	Mano and Ciletti
Publisher/Year:	Pearson/2018

Reference Materials	
Title:	Tutorial on Sequential Circuit Design
Author:	D. Capson
Publisher/Year:	latest version will be provided on course *Brightspace* website
Title:	Tutorial on Memory Systems
Author:	D. Capson
Publisher/Year:	latest version will be provided on course *Brightspace* website

Course Delivery

Lectures will be given primarily Face-to-Face (F2F) in the designated lecture room on campus as scheduled. Occasionally, some course lecture/tutorial content may be delivered on-line (using Zoom) and may include preparation for midterm tests, labs, and the Final exam.

All labs will be completed Face-to-Face (F2F) in the designated laboratory room on campus. Each student is required to submit a printed copy of an individually completed pre-Lab prior to the scheduled start time of each of their labs. Each lab group will have a maximum of 2 students. The laboratory grading rubric document will be found on the course *Brightspace* page.

All tests will be completed Face-to-Face (F2F) in a designated room on campus. All tests and the Final Exam will be closed book - no notes, textbooks, laptops/tablets, or smartphones are permitted.

Tutorials will be integrated with lectures and conducted Face-to-Face (F2F) during scheduled lecture slots.

Office hours for consulting with the instructor can be provided either on-line (via Zoom) or be in-person office meetings. Time slots for office hours will be arranged individually by contact with the instructor via e-mail.

The course *Brightspace* page will be used to distribute all course materials, including labs.

Learning & Teaching Technologies

Recommended tools:

- Desktop or Laptop Computer with webcam/microphone/speakers/headsets and Zoom software for online lectures, tutorials and instructor office hours (if and when necessary). Please make yourself familiar with Zoom.

Assessment

Assessment Tool	Weight	Date
Mid-term #1	15%	Friday, October 9, 2026 (tentatively)
Mid-term #2	15%	Wednesday, November 4, 2026 (tentatively)
Labs	30%	8 labs, as scheduled
Final Exam	40%	TBD

Note: There will be no assignments required to be submitted for credit. Throughout the course, suggested problem sets will be specified on the course *Brightspace* page, but these will be neither corrected nor graded. Solutions to all suggested problems will be available via the course Brightspace page.

Notes

- Failure to attend all Laboratories will result in a grade of N.
- Failure to complete all Laboratory requirements will result in a grade of N
- You must pass the final exam (i.e. at least 50%) to pass this course.

- The final grade obtained from the above marking scheme for the purpose of GPA calculation will be based on the percentage-to-grade point conversion table as listed in the current Undergraduate Calendar.
- Coursework Mark Appeals: All marks must be appealed within 7 days of the mark being posted.

- A student who has written the final exam and received a failing course grade may be offered an opportunity to write a supplemental exam. The percentage grade at which the instructor recommends supplemental exams varies, depending on the methods of evaluation for the course. Supplemental exam recommendations are decided by the department and the faculty based on a review of each student's overall academic record.

Additional Notes

- The instructor's policy on the use of AI tools (including, but not limited to, ChatGPT) will be found on the course *Brightspace* page. The University's AI Position Statement may be found at: https://teachanywhere.uvic.ca/academic-integrity/ai-evaluation/.

General Information

Note to students: Students who have issues with the conduct of the course should discuss them with the instructor first. If these discussions do not resolve the issue, then students should feel free to contact the Chair of the Department by email, or the Chair's Assistant to set up an appointment.

Course Lecture Notes

Unless otherwise noted, all course materials supplied to students in this course have been prepared by the instructor and are intended for use in this course only. These materials are not to be recirculated, whether by emailing, uploading, copying to websites, or to others not enrolled in this course. Violation of this policy may in some cases constitute a breach of academic integrity as defined in the UVic Calendar.

Academic Integrity

Academic Integrity requires commitment to the values of honesty, trust, fairness, respect, and responsibility. It is expected that students, faculty members and staff at the University of Victoria, as members of an intellectual community, will adhere to these ethical values in all activities related to learning, teaching, research and service. Any action that contravenes this standard, including misrepresentation, falsification or deception, undermines the intention and worth of scholarly work and violates the fundamental academic rights of members of our community. This policy is designed to ensure that the university's standards are upheld in a fair and transparent fashion.

Attendance

Students are expected to attend all classes in which they are enrolled. An academic unit may require a student to withdraw from a course if the student is registered in another course that occurs at the same time.

- An instructor may refuse a student admission to a lecture, laboratory, online course discussion or learning activity, tutorial or other learning activity set out in the course outline because of lateness, misconduct, inattention or failure to meet the responsibilities of the course set out in the course outline. Students who neglect their academic work may be assigned a final grade of N or debarred from final examinations.
- Students who do not attend classes must not assume that they have been dropped from the course by an academic unit or an instructor. Courses that are not formally dropped will be given a failing grade; students may be required to withdraw and will be required to pay the tuition fee for the course.

See further information on attendance in the Undergraduate Academic Calendar.

Centre for Accessible Learning (CAL)

The University of Victoria is committed to creating a learning experience that is as accessible as possible. If you are registered with the Centre for Accessible Learning and anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with me. If you are a student with a disability or chronic health condition, you can meet with a CAL advisor to discuss access and accommodations.

Academic Concession

The University recognizes its responsibility to offer academic concessions to students whose ability to complete course requirements is interrupted by unexpected and unavoidable circumstances or conflicting responsibilities. Review the Academic Concession Regulation in the Undergraduate Academic Calendar and website.

Grading System

The University of Victoria follows a percentage grading system in which the instructor will submit grades in percentages. The University will use the following Senate approved standardized grading scale to assign letter grades. Both the percentage mark and the letter grade will be recorded on the academic record and transcripts.

F	D	C	C+	B-	B	B+	A-	A	A+
0-49	50-59	60-64	65-69	70-72	73-76	77-79	80-84	85-89	90-100

Grades	Description
A+, A, A-	**Exceptional**, **outstanding** or **excellent** performance. Normally achieved by a minority of students. These grades indicate a student who is *self-initiating*, *exceeds expectation* and has an *insightful* grasp of the subject matter.
B+, B, B-	**Very good**, **good** or **solid** performance. Normally achieved by the largest number of students. These grades indicate a *good* grasp of the subject matter or *excellent grasp in one area balanced with satisfactory grasp in the other areas*.
C+, C	**Satisfactory**, or **minimally satisfactory**. These grades indicate a *satisfactory performance and knowledge* of the subject matter.
D	**Marginal Performance**. A student receiving this grade demonstrated a *superficial grasp* of the subject matter.
F	**Unsatisfactory performance**. Wrote final examination and completed course requirements; no supplemental.

Student Experience of Learning (SEL)

I value your feedback on this course. Towards the end of term you will have the opportunity to complete a confidential Student Experience of Learning (SEL) survey regarding your learning experience. The survey is vital to providing feedback to me regarding the course and my teaching, as well as to help the department improve the overall program for students in the future. When it is time for you to complete the survey, you will receive an email inviting you to do so. If you do not receive an email invitation, you can go directly to the SEL site

You will need to use your UVic NetLink ID to access the survey, which can be done on your laptop, tablet or mobile device. I will remind you closer to the time, but please be thinking about this important activity, especially the following three questions, during the course.

- What strengths did your instructor demonstrate that helped you learn in this course?
- Please provide specific suggestions as to how the instructor could have helped you learn more effectively.
- Please provide specific suggestions as to how this course could be improved.

Equality

This course aims to provide equal opportunities and access for all students to enjoy the benefits and privileges of the class and its curriculum and to meet the syllabus requirements. Reasonable and appropriate accommodation will be made available to students with documented disabilities (physical, mental, learning) in order to give them the opportunity to successfully meet the essential requirements of the course. The accommodation will not alter academic standards or learning outcomes, although the student may be allowed to demonstrate knowledge and skills in a different way. It is not necessary for you to reveal your disability and/or confidential medical information to the course instructor. If you believe that you may require accommodation, the course instructor can provide you with information about confidential resources on campus that can assist you in arranging for appropriate accommodation. Alternatively, you may want to contact the Centre for Accessible Learning located in the Campus Services Building.

The University of Victoria is committed to promoting, providing, and protecting a positive, and supportive and safe learning and working environment for all its members.

Resources for students

a. UVicLearn is the primary learning resource for students that offers many learning workshops and resources to help students with academics and learning strategies.
b. Library resources
c. Indigenous Student Services (ISS)
d. Academic Skills Centre (ASC)
e. Math & Stats Assistance Centre (MSAC)
f. Learning Assistance Program (LAP)
g. Community-Engaged Learning (CEL)
h. Learning in Motion - Co-op and Career Portal
i. Academic concession
j. Academic accommodation
k. Academic accommodation & access for students with disabilities
l. Student groups and resources
m. Student wellness
n. Ombudsperson

University statements and policies

a. University Calendar - Information for all students
b. Accommodation of Religious Observance
c. Student Conduct
d. Non-academic Student Misconduct
e. Academic Accommodations and Accessibility
f. Diversity / EDI – VPAC commitment
g. Sexualized Violence Prevention and Response
h. Discrimination and Harassment

Copyright Statement

All course content and materials are made available by instructors for educational purposes and for the exclusive use of students registered in their class. The material is protected under copyright law, even if not marked with a ©. Any further use or distribution of materials to others requires the written permission of the instructor, except under fair dealing or another exception in the Copyright Act. Violations may result in disciplinary action under the Resolution of Non-Academic Misconduct Allegations policy (AC1300).